

03_gradio_interface

September 27, 2026

```
[1]: import gradio as gr
```

0.0.1 Remember to close the port with `.close()` or by restarting the kernel. If you don't, it sometimes struggles to re-run the code cause the port is busy.

```
[2]: def create_app():
    with gr.Blocks() as interface:
        gr.Markdown(r"""
            # Spoken Digit Classification
            ### Project description
            This project uses the [Audio MNIST Dataset](https://huggingface.co/
            ↪datasets/gilkeyio/AudioMNIST) to train a model that predicts numbers from 0_
            ↪to 9 based on audio input. The neural network is a multilayer perceptron_
            ↪with an input of 26 neurons, two hidden layers of 64 and 32 neurons, and a_
            ↪final output layer of 10 neurons. It uses the Adam optimizer and the_
            ↪cross-entropy loss function.
            """)

        with gr.Tabs():
            # ===== DATA EXPLORATION TAB_
            ↪=====

            with gr.Tab("Data"):
                with gr.Group():
                    gr.Markdown("#### Dataset Exploration")

                    gr.Markdown("#### *Dummy data for testing purposes, review_
                    ↪and update*")

                    gr.Markdown("""
                        The Audio MNIST dataset contains 3,000 audio recordings of_
                        ↪spoken digits (0-9) from 60 different speakers.

                        Each recording is approximately 1 second long and sampled_
                        ↪at 8 kHz. The dataset is split into training (2,400 samples)
                        and testing (600 samples) sets with a balanced distribution_
                        ↪across all digit classes.
                        """)
```

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gr.Markdown("**Dataset Statistics:**")

# Dummy table data
table_data = [
    ["Total Samples", "3000"],
    ["Training Samples", "2400"],
    ["Testing Samples", "600"],
    ["Number of Classes", "10 (digits 0-9)"],
    ["Sample Rate", "8 kHz"],
    ["Average Duration", "1.0 seconds"],
    ["Speakers", "60"],
]

gr.Dataframe(
    value=table_data,
    headers=["Metric", "Value"],
    interactive=False,
    label="Dataset Overview"
)

# ===== TRAINING INTERFACE TAB
↪=====
with gr.Tab("Train"):
    gr.Markdown("#### Training Interface")

# ===== MODEL EVALUATION TAB
↪=====
with gr.Tab("Model"):
    gr.Markdown("#### Model Evaluation")

gr.Markdown(r"""
    *Created by: Amets Martiarena, Teva Philippe, Alvaro Crespo | Check
    ↪the original repo [here] (https://github.com/AmetsMarti/SDOML-project1)*.
    """)
return interface

```

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[3]: try:
    interface.close()
except:
    pass

# Set to true to publish the app
send_it = False

interface = create_app()
interface.launch(server_port=7860, share=send_it)

```

```
* Running on local URL:  http://127.0.0.1:7860
* To create a public link, set `share=True` in `launch()`.

<IPython.core.display.HTML object>
```

[3]:

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